

Bioinformatics assignment

Code Alpha

Task no.1: DNA/Protein Sequence Analysis

Step1:

UniProtKB/Swiss-Prot: P68871.2

[GenPept](#) [Identical Proteins](#) [Graphics](#)

```
>sp|P68871.2|HBB_HUMAN RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin;
AltName: Full=Hemoglobin beta chain; Contains: RecName: Full=LVV-hemorphin-7; Contains:
RecName: Full=Spinorphin
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKVKAHGKKVLG
AFSDGLAHLDNLKGTATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVAN
ALAHKYH
```

Step 2:

i Your search is limited to records **exclude: non-redundant RefSeq proteins**

Job Title	sp P68871.2		
RID	3CJVA7WS014	Search expires on 06-21 16:12 pm	Download All ▼
Program	BLASTP 	Citation ▼	
Database	ClusteredNR	See details ▼	
Query ID	P68871.2		
Description	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-		
Molecule type	amino acid		
Query Length	147		
Other reports	Distance tree of results	Multiple alignment	MSA viewer 

☒	1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	263	263	87%	9e-88	100.00%	175	AAA35952.1
☒	1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	277	277	100%	2e-93	93.20%	147	AAA35597.1
☒	1 member(s), 1 organism(s)	giant panda	hypothetical protein PANDA_015769 [Ailuropoda melanoleu...	275	446	100%	3e-91	90.48%	258	EFB18430.1
☒	1 member(s), 1 organism(s)	rabbit	hemoglobin subunit beta-1/2 isoform 1x [Oryctolagus cunicu...	253	253	100%	8e-84	90.48%	170	NP_001300972.1
☒	7 member(s), 6 organism(s)	dog_coyote_wolf_fox	hemoglobin subunit beta-like [Canis lupus familiaris]	275	275	100%	1e-92	89.80%	147	NP_001257812.1
☒	185 member(s), 76 organism(s)	placentals	hemoglobin subunit delta [Ailuropoda melanoleuca]	273	273	100%	6e-92	89.80%	147	NP_001291814.1
☒	19 member(s), 18 organism(s)	placentals	hemoglobin subunit beta-like [Ailuropoda melanoleuca]	273	273	100%	7e-92	89.80%	147	NP_001291812.1
☒	5 member(s), 5 organism(s)	Old World monkeys	PREDICTED: LOW QUALITY PROTEIN: hemoglobin subun...	270	270	100%	8e-91	89.80%	147	XP_011830554.1
☒	9 member(s), 6 organism(s)	bats	hemoglobin subunit beta [Rhynchonycteris naso]	271	271	100%	2e-91	89.12%	147	KAM8815943.1
☒	1 member(s), 1 organism(s)	black flying fox	Hemoglobin subunit beta [Pteropus alecto]	272	518	100%	2e-88	89.12%	363	ELK09417.1

Analysis:

The table documents several key homologous sequences across different mammalian groups:

- Human (*Homo sapiens*): The top hits are for beta-globin, with one sequence showing 100% identity over 100% coverage, confirming a perfect match to the human reference.
- Giant Panda (*Ailuropoda melanoleuca*): Shows high homology with a 90.48% identity and an E-value of $3e^{-91}$, suggesting a very conserved hypothetical protein related to hemoglobin.
- Rabbit (*Oryctolagus cuniculus*): A sequence for hemoglobin subunit beta-1/2 also shows 90.48% identity, further illustrating the conservation of this protein in mammals.
- Other Species: Significant matches are also found in dogs, wolves, foxes, Old World monkeys, bats, and black flying foxes, with identities consistently around the 89% to 90% range.

Explanation of all the core metrics in table:

1. Query Coverage: This indicates how much of your input sequence is covered by the match. Most results in the table show 100% Query Cover, meaning the entire query sequence was successfully aligned with the subject sequence in the database.
2. E-value (Expect Value): This is the most critical statistical metric. Values in the table such as $2e^{-93}$ or $1e^{-92}$ are essentially zero, which indicates that the matches are highly significant and extremely unlikely to have occurred by chance.
3. Per. Ident (Percentage Identity): This shows how many amino acids are identical between the two sequences. High percentages (ranging from 89.12% to 100%) confirm that these sequences are closely related homologs.